Incremental Predictive Value of Two Antenatal Visits for Recorded Low Birth Weight

Technical Report
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Abstract

This study tests whether measurements from the first antenatal visit improve prediction of recorded low birth weight beyond the second visit alone, using a public Tanzanian antenatal dataset of 8,817 records. From an analysis cohort of 3,908 pregnancies with 206 low birth weight births, a held out partition of 782 pregnancies with 41 events was sealed before any model was fitted. Three nested feature sets were compared on the 3,126 development pregnancies (165 events) with repeated nested grouped cross validation, average precision as the primary measure and group bootstrap intervals. Adding the first visit did not improve prediction by any amount the design could detect: the logistic difference was −0.0040 (95 percent interval −0.0130 to +0.0025) against a smallest detectable difference of 0.0110. Adding the second visit to registration data gave a modest gain in discrimination. None of 28 prespecified improvement experiments passed the replacement rule fixed in advance. The single held out evaluation agreed with the development results, with weaker overall discrimination and wide intervals. A prototype application serves the frozen model on synthetic cases only.

Introduction

Routine antenatal records in East Africa capture a small set of measurements at each visit. Whether an earlier visit adds predictive information to a later one matters for how community health workers prioritise follow up, and it had not been tested on open data with the test records kept apart. The proposal fixed four research questions before any model was fitted:

1. Do measurements from the first antenatal visit improve prediction of recorded low birth weight beyond the second visit alone in the eligible cohort?
2. Does a learned model outperform constant prevalence prediction, and does adding the earlier visit improve both discrimination and calibration over the second visit features alone?

3. Within fixed ranking budgets, what proportion of pregnancies recorded as low birth weight is retrieved by model ranking relative to ranking at random?
4. What sensitivity do the feature set comparisons show to missing measurements and to questionable outcome records?

The proposal also fixed the cohort, the partition, the feature sets, the evaluation and a prototype. Development results are internal validation. The held out partition was evaluated once, on 24 October 2026, on models locked beforehand.

Methods

Data and cohort

The released file is read only after its SHA 256 matches a frozen value, and it is never modified. Pure parsers turn text cells into typed values: blood pressure through a regular expression, sentinel tokens such as "not_checked" into missing values, and unambiguous numeric slips repaired. Values outside the physiological ranges held in one frozen study specification become missing. A cohort builder applies one named method per eligibility step and records the reason for every excluded record, giving an eligible cohort of 8,161 and an analysis cohort of 3,908 pregnancies with 206 low birth weight births.

Partition and leakage control

Records that share a registration profile (age, last menstrual date, height, registration weight and prior deliveries) form a group, and whole groups are assigned to folds while balancing events. The first fold is the held out partition of 782 pregnancies with 41 events, written with a checksum before any model step. Every model fit carries a training guard that refuses any held out record, and held out outcomes are read only through one logged gate. The development partition holds 3,126 pregnancies with 165 events. An integrity audit of 11 checks passed before and after the held out evaluation.

Feature sets

Set A holds registration characteristics. Set B adds the visit two measurements. Set C adds the visit one value of each measurement recorded at both visits for at least 60 percent of records, a screen refitted in every training fold. Linear models receive values only; tree models also receive the change between visits. The sets are subclasses of one abstract feature set class.

Models and validation

Eight model families and a constant prevalence baseline were compared. Pipelines impute

medians with missingness indicators and scale where needed; gradient boosted trees handle missing values natively. The design is 5 outer folds repeated 10 times, with hyperparameters chosen in 3 grouped inner folds by average precision, the area under the precision recall curve. Predictions are averaged per pregnancy across repeats, and paired differences between feature sets on the same pregnancies receive a group bootstrap interval from 2,000 resamples. The smallest detectable difference is 2.8 bootstrap standard errors, corresponding to a five percent two sided test at 80 percent power. Sensitivity analyses S1 to S6 and multiple imputation test the primary comparison under changed rules. Stage 2 ran 28 prespecified improvement experiments under a replacement rule written before any of them ran.

Prototype

A FastAPI application with routers that validate, services that decide, and a database of synthetic cases serves the model. Scoring goes only through the frozen artefact, whose checksum is verified at start; a changed file stops the server. A Next.js interface shows rank, probability with an interval, and the label "Likely" or "Not likely low birth weight" at the threshold fixed at the highest 20 percent of out of fold predictions, with how often that label is right. Sign in with administrator and researcher roles is required, and the application is hosted online with a local backup.

Implementation and verification

Scope alignment

Planned component	Status	Evidence
Data loader with checksum; cleaning and eligibility (R1)	Completed	outputs/tables/cohort_flow.csv
Partition writer (R3) and guard	Completed; integrity audit 11 of 11	docs/evidence/holdout_integrity.md
Feature sets A, B, C (R2) and nested cross validation	Completed	outputs/tables/feature_set_comparisons.csv
Sensitivity analyses S1 to S6 and extensions	Completed	outputs/tables/sensitivity_analyses.csv
Prespecified improvement experiments (Stage 2)	Completed: 28 runs, 0 passed	configs/experiment_plan.yaml, outputs/tables/experiments_v2.csv
Statistical analysis of the research questions	Completed	notebooks/03_statistical_analysis.ipynb, docs/research_results.md
Held out evaluation (R5)	Completed once, logged	docs/holdout_results.md, partitions/holdout_access.log

Planned component	Status	Evidence
Model serialiser (R12, R16) and label (decision D6)	Completed	webapp/model/version.json, webapp/model/classification.json
Data model (Section 3.4.4)	Completed in SQLite; the budget named PostgreSQL	python -m antenatal load-db
Interface screens 1 to 5, entry rules (R9, R10), sign in (D1)	Completed in Next.js over the API	frontend/src/, docs/evidence/input_tests.md
Cross browser, accessibility, performance, container, online host	Completed; hosted online with local backup (D7)	docs/evidence/, Dockerfile, fly.toml

Testing

The test suite ran 329 tests in 336.4 seconds with 0 failures: 253 unit tests, 70 integration tests and 6 browser tests. Line coverage of the package and the application was 96.3 percent of 4,435 statements. Ruff reported no findings and black reported no changes needed. Unit tests cover the parsers on odd values from the file, each cohort count, partition disjointness, sealing and tamper detection, and each metric against scikit-learn or a hand computation. Leakage and integrity tests confirm that no held out record appears in any saved prediction, that the guard refuses held out records, and that the served model rebuilds from development records to within 1e-9. Integration tests of the API cover every refusal path, sign in and roles, a tampered artefact refusing to start, and equality of API, service and package to within 1e-9. End to end tests run the full command sequence on a throwaway project, and browser tests exercise the five screens in three browsers at two sizes.

An input variation run entered 12 inputs through a running server and 12 of 12 behaved as expected. A typical complete case, age 55 at the upper boundary, visit two at 28 weeks at the upper boundary, and visit two with pulse and haemoglobin missing each returned 201 with an assessment. A missing synthetic flag, age 121, missing height, visit one systolic 300, visit one diastolic 20, visit one at 30 weeks, and visit two at the same week as visit one each returned 422 with the expected message. Visit two entered before visit one returned 409 with "Record visit one before visit two".

Code quality

The package separates data access, parsing, cohort rules, partitioning, features, models, evaluation, the artefact and reporting into their own modules, and every constant lives in one frozen study specification. The feature set and metric strategy classes use abstraction and polymorphism. Every public class and function has a docstring and type hints; no function exceeds 50 lines and no file 300. Ruff and black run locally and in continuous integration, every output carries a checksum in a manifest, the README results are generated and tested, and the

history holds 94 commits with imperative messages.

Results

Result	Value (95 percent interval)
Primary, development: first visit added (C over B), average precision, logistic	−0.0040 (−0.0130 to +0.0025)
Primary, development: same, trees	−0.0066 (−0.0173 to +0.0024)
Smallest detectable difference, C over B, logistic	0.0110
Secondary, development: second visit added (B over A), ROC AUC, logistic	+0.0205 (−0.0066 to +0.0476)
Secondary, development: same, trees	+0.0345 (+0.0070 to +0.0627)
Sensitivity analyses with a C over B interval excluding zero	0 of 7
C over B, pooled multiple imputation	−0.0055 (−0.0151 to +0.0008)
Calibration slope, logistic, set C, development	0.98 (0.63 to 1.37)
Share of events in the top 10, 20 and 30 percent, development	21.8 percent (15.9 to 27.9); 34.5 percent (27.8 to 42.0); 47.9 percent (40.4 to 55.4)
Stage 2: largest average precision gain (weighted gradient boosting on C)	+0.0075 (−0.0036 to +0.0243); detectable 0.0193
Held out: primary, C over B, average precision	+0.0012 (−0.0221 to +0.0185), p 0.707
Held out: secondary, B over A, ROC AUC	+0.0405 (+0.0008 to +0.0805), p 0.046
Held out: served model average precision; ROC AUC	0.1023 (0.0532 to 0.1879); 0.5725 (0.4752 to 0.6721)
Held out: calibration slope of the served model	0.4475 (−0.4110 to 1.4021)
Held out: label sensitivity; specificity; correct when "Likely"	0.3415 (0.2059 to 0.4865); 0.7949 (0.7643 to 0.8226); 0.0843 (0.0471 to 0.1312)

Discussion

On research question 1, adding the first visit to the second did not improve prediction of recorded low birth weight by any amount this design could detect. The result holds on development records in both model families, in all seven sensitivity analyses, under multiple imputation, and in the single held out evaluation. On research question 2, the learned models beat the prevalence baseline on development records, and the second visit over registration alone shows a modest gain in discrimination that reached significance for trees on development records and for the locked comparators on the held out records (p 0.046). Calibration of the served model was good on development records (slope 0.98) and uncertain on the held out records. On